स्री डैक
CDAC
Section 2 of 2

Question # 1

Revisit

Suppose the distribution of salaries in a company X has median $35,000, and 25th and 75th percentiles are $21,000 and $53,000 respectively. Would a person with Salary $1 be considered an Outlier?

Choose the best option

- Yes
- No
- More information is required
- none of these choices

Question # 2

Revisit

In performing a hypothesis test where the null hypothesis is that the population mean is 45 against the alternative hypothesis that the population mean is not equal to 45, a random sample of 17 items is selected. The sample mean is 47.6 and the sample standard deviation is 3.3. It can be assumed that the population is normally distributed. The degrees of freedom associated with this are:

Choose the best option

- 2
- 15
- 16
- 17

Question # 5

Revisit

In performing a hypothesis test where the null hypothesis is that the population mean is 6.9 against the alternative hypothesis that the population mean is not equal to 6.9, a random sample of 16 items is selected. The sample mean is 7.1 and the sample standard deviation is 2.4. It can be assumed that the population is normally distributed. The observed "t" value for this problem is:

Choose the best option

- 0.05
- 0.2
- 0.33
- 1.33

Question # 4

Revisit

A simulation uses the logical relationships and mathematical expressions of the

Choose the best option

- Real system
- Computer model
- Performance measures
- Inferences

Question # 3

Revisit

In simple linear regression, which of the following statements indicates there is no linear relationship between the variables x and y?

Choose the best option

- Coefficient of determination is −1.0.
- Coefficient of correlation is 0.0.
- Sum of squares for error is 0.0.
- none of these choices

Question # 6

Revisit

In testing hypotheses, the researcher initially assumes:

Choose the best option

- the alternative hypothesis is true.
- the null hypothesis is true
- errors cannot be made
- the population parameter of interest is known

Question # 7

Revisit

Whenever hypotheses are established such that the alternative hypothesis is ">", then this would be a ________.

Choose the best option

- ~~two-tailed test~~
- Type II test
- one-tailed test
- Type I test

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

- [illegible]
- [illegible]
- [illegible]
- [illegible]

[illegible]

Question # 8

Revisit

The previous probabilities in Bay's Theorem that are changed with the help of new available information are classified as

Choose the best option

- Independent probabilities
- Posterior probabilities
- Interior probabilities
- Dependent probabilities

सी डैक
CDAC
Section 2 of 2

PG-DBDA_0921_220322

Total 00:58:12
Section 00:58:12
Finish Test

Question # 12

Revisit

For a given level of significance, if the sample size is increased, the power of the test will

Choose the best option

- ○ increase
- ○ stay the same
- ○ decrease
- ◉ none

Clear Response

[illegible]

[illegible]

[illegible]

[illegible]

- [illegible]
- [illegible]
- [illegible]
- [illegible]

[illegible]

Question # 15

Revisit

When a true null hypothesis is rejected, the researcher has committed a:

Choose the best option

- ○ Type II error
- ◉ Type I error
- ○ rejection error
- ○ error in judgment

Clear Response

Question # 14

Revisit

A statistics professor asked students in a class their ages. On the basis of this information, the professor states that the average age of all the students in the university is 21 years. This is an example of

Choose the best option

- ○ census
- ○ descriptive statistics
- ○ an experiment
- ○ statistical inference

Question # 13

Revisit

The standard deviation of a sample of 100 observations equals 64. The variance of the sample equals

Choose the best option

- 8
- 10
- 6400
- 4096

Clear Response

64×64

σ

$[\sigma = V^2]$

$V = \sum_{n=1} x_i + y_i$

Question # 18

Revisit

The expected value of Y is a linear function of the X(X1,X2....Xn) variables and regression line is defined as: Y = β0 + β1 X1 + β2 X2......+ βn Xn. Which of the following statement(s) are true? 1) If Xi changes by an amount ΔXi, holding other variables constant, then the expected value of Y changes by a proportional amount βi ΔXi, for some constant βi (which in general could be a positive or negative number). 2) The value of βi is always the same, regardless of values of the other X's. 3) The total effect of the X's on the expected value of Y is the sum of their separate effects. Note: Features are independent of each others(zero interaction).

Choose the best option

- 1 and 2
- 1 and 3
- 2 and 3
- 1,2 and 3

Question # 17

Revisit

In simple linear regression, the coefficient of correlation r and the least squares estimate b1 of the population slope β1:

Choose the best option

- ○ must be equal.
- ○ must have the same sign.
- ○ are not related.
- ◉ none of these choices

Clear Response

Question # 16

Revisit

Generally, which of the following method(s) is used for predicting continuous dependent variable?
1. Linear Regression
2. Logistic Regression

Choose the best option

- ○ 1 and 2
- ○ only 1
- ○ only 2
- ◉ none of these choices

Clear Response

Question # 27

Revisit

Which of the following indicates a fairly strong relationship between X and Y?

Choose the best option

- Correlation coefficient = 0.9
- The p-value for the null hypothesis Beta coefficient =0 is 0.0001
- The t-statistic for the null hypothesis Beta coefficient=0 is 30
- none of these choices

Clear Response

CDAC
Section 2 of 2

PG-DBDA_0921_220322

18 19 20 21 22 23 24 25 26 27 28 29 30 < 26 of 40 >

Question # 26

Which of the following criteria cannot be used to determine the number of factors in an EFA?

Revisit

Choose the best option

- ○ Asking a group of researchers before the analysis
- ○ Eigenvalue rule
- ○ Scree test
- ◉ Parallel analysis

Clear Response

Question # 19

Revisit

The strength of the linear relationship between to numerical variables may be measured by the

Choose the best option

- scatter diagram
- ~~coefficient~~ of correlation
- slope
- Y intercept

CDAC

Section 2 of 2

PG-DBDA_0921_220322

23 24 25 26 27 28 29 30 31 32 33 34 35 < 31 of 40 >

Question # 31

Which of the following is Step 1 for Performing Simulation Analysis

Revisit

Choose the best option

- ○ choose input variables
- ○ create entities for the simulation process
- ○ prepare a problem statement
- ○ determine the output variables

Question # 29

Revisit

Whenever hypotheses are established such that the alternative hypothesis is "not equal to", then the test is a:

Choose the best option

- Type I test
- two-tailed test
- one-tailed test
- Type II test

CDAC

Section 2 of 2

PG-DBDA_0921_220322

18 19 20 21 22 23 24 25 26 27 28 29 30 < 28 of 40 >

Question # 28

What will a factor loading in an orthogonal solution represent?

Revisit

Choose the best option

- ○ Correlation
- ○ Partial correlation
- ○ Multiple correlation
- ◉ Eigenvalue

Clear Response

सी-डैक CDAC

Section 2 of 2

PG-DBDA_0921_220322

23 24 25 26 27 28 29 30 31 32 33 34 35 < 33 of 40 >

Question # 33

Revisit

The statistical distribution used for testing the difference between two population variances is the ____ distribution.

Choose the best option

- ○ t
- ○ standardized normal
- ○ binomial
- ◉ F

Clear Response

CDAC

Section 2 of 2

PG-DBDA_0921_220322

23 24 25 26 27 28 29 30 31 32 33 34 35 < 32 of 40 >

Question # 32

Revisit

Problems which seek to maximise or, minimise profit or, cost form a general class of problems called

Choose the best option

- Simple problems
- Difficult problems
- Optimisation problems
- Non-linear problems